

VI Semester B.E. Examination
(Electronics and Communication Engineering)
Automotive Electronics (22EECC305)

Duration: 3 hours

Max. Marks: 100

Note: i) Answer any TWO full questions from UNIT-I, any TWO full questions from UNIT-II and any ONE full question from UNIT-III.

ii) Use of steam table is permitted

iii) Assume suitable data if necessary with justification

UNIT-I

- 1 a. Define stoichiometric ratio. Calculate λ for AFR = 12.2. Derive fuel injector pulse duration (T_w) for open- and closed-loop modes using the speed-density method. (08marks)
- b. At 8000 rpm, driver demands more speed. Explain how ECU adjusts ignition timing and propose instrumentation for closed-loop ignition control. (06marks)
- c. Explain the need for EGO Sensor. Draw the schematic of EGO Sensor and analyze the output for rich and lean mixtures. (06marks)
- 2 a. Calculate the injector pulse duration and fuel quantity for an 8-cylinder engine at 4000 rpm (a) fully warmed-up and (b) very cold, given fuel flow = 0.0012 kg/s and mass air flow = 0.0025 kg/s. Explain PI/PID use in engine management. (08marks)
- b. Why idle speed control is necessary for an engine? Explain its configuration with a necessary block diagram. (06marks)
- c. For the development of an Engine ECU, apply MBD approach adhering to automotive V design model. (06marks)
- 3 a. Choose between magnetic reluctance and Hall Effect sensors for engine speed measurement. Justify and give a schematic. (08marks)
- b. In closed-loop mode, explain how changes in (i) EGR, (ii) AFR, and (iii) ignition timing affect engine performance, with plots. (06marks)
- c. Discuss how secondary air injection system regulates air supply to EMS with respect to different operating conditions? (06marks)

UNIT-II

- 4 a. Describe causes of wheel lock and slide during braking, and how ABS prevents them. Calculate slip % for a car at 60 mph, wheel radius 16 in, wheel speed 2000 rpm. (08marks)
- b. What is cruise controller? What are the disadvantages of it and how are they overcome, Explain in details. (06marks)
- c. Explain the principles, working, and design of an automotive airbag system, used for passenger safety during various collisions. (06marks)
- 5 a. Answer the following with respect to CAN communication
- i) Classical CAN vs CAN FD
- ii) CAN Frame format (08marks)
- b. Describe the control system which provides a solution for wheel spinning, and discuss the related control functions. (06marks)
- c. Define oversteer and understeer. Using brake circuit configuration, show how ESP corrects these conditions. (06marks)
- 6 a. Compare event driven and time triggered communication strategies. Discuss with the scenario how Flexray incorporates both of these strategies into a common protocol. (08marks)
- b. Illustrate the role of electronic suspension system for ride and handling by controlling the damping coefficient. (06marks)

- c. Calculate nominal and maximum T Header, T Response, and T Frame, if LIN is operating at 10 Kbps baud rate and reserved time is set to 30% for transmitting four bytes of data. (06marks)

UNIT-III

- 7 a. Describe ISO 26262. For an ESCL system, perform hazard analysis and ASIL-based risk assessment with likelihood estimation. (10marks)
- b. List key ADAS features in modern vehicles and explain any two of them that enhance safety and driver assistance. (10marks)
- 8 a. Discuss OBD objectives. For a 2022 Audi A8 with rough running and engine light on, list possible faults. (10marks)
- b. During cold cranking the air/fuel ratio is not able to be controlled. List the possible faults. Discuss the features of diagnostic protocol KWP2000. (10marks)